CHAPTER SIX

RESULTS AND DISCUSSION

6.0 Introduction

This chapter presents the findings obtained from the design, implementation, and evaluation of the proposed Mobile Application for Hall Booking and Event Management at Jazeera University. It highlights how the system performed in relation to the research objectives and evaluates its effectiveness in addressing the limitations of the existing manual processes.

The chapter first outlines the results of functional testing, usability assessment, and user feedback to demonstrate how well the application meets requirements such as real-time hall availability, booking requests, and push notifications. It then discusses the implications of these results by comparing them with findings from related studies and theoretical models.

Furthermore, the discussion interprets the significance of the outcomes in improving efficiency, user experience, and institutional resource management. Strengths, limitations, and areas of improvement are also analyzed, providing a balanced view of the system's overall performance.

6.1 Results Presentation

The implementation of the Jazeera University Hall Event App produced tangible results that directly addressed the issues identified in the problem statement. The system was tested using several scenarios including user authentication, booking, event management, and role-based access control.

6.2 Research questions answers

1. What are the key problems in the current manual hall booking process at Jazeera University?

The current manual hall booking process at Jazeera University faces multiple operational and administrative challenges that hinder efficiency and transparency. The system relies heavily on paperwork and physical interactions between students, staff, and administrators. This method is time-consuming and prone to human error. For instance, double bookings often occur when two departments or student organizations unknowingly reserve the same hall for overlapping times.

Additionally, there is no centralized database to track hall usage or booking history, which makes it difficult to manage space allocation and generate reports for planning purposes. Approvals are typically handled manually, leading to delays and miscommunication between users and the administration.

The lack of real-time updates further worsens the issue, as staff and students cannot instantly verify whether a hall is available before making a request. This results in frustration, inefficiency, and scheduling conflicts. Overall, the absence of automation, poor coordination, and lack of transparency make the current manual system unsuitable for the university's growing academic and extracurricular needs.

2. What challenges do students and staff face when using the manual system?

Students and staff encounter several difficulties when dealing with the manual hall booking system. First, accessibility is limited — users must physically visit administrative offices to check hall availability or submit booking requests, which is inconvenient, especially for those with tight academic schedules.

Second, communication between stakeholders is inefficient. Because there is no notification or tracking system, users are often unaware of the status of their booking requests until they personally follow up. This uncertainty leads to confusion and sometimes the cancellation or postponement of important events.

Third, manual record-keeping makes it difficult to retrieve historical booking data, which administrators could otherwise use to plan future events or analyze hall utilization. Students also experience frustration due to the lack of transparency in approval processes and the frequent misplacement of booking forms.

Finally, because the process is entirely paper-based, it consumes unnecessary resources such as printing materials and administrative time. These challenges collectively highlight the urgent need for a digital, automated solution that ensures convenience, reliability, and accountability for all users.

3. How can a mobile-based system improve efficiency and user satisfaction?

A mobile-based hall booking system can significantly transform how Jazeera University manages events and facilities. By integrating cloud computing and mobile technologies, the system offers real-time access to hall availability, allowing users to book, modify, or cancel reservations instantly.

Automation eliminates manual paperwork and reduces administrative workload by providing digital approval workflows, where notifications are automatically sent to users once a booking is approved or rejected. This ensures faster decision-making and enhances communication between students, staff, and administrators.

Moreover, the mobile-based system offers convenience and flexibility. Users can access the platform anytime and anywhere, making it easier to manage bookings without visiting administrative offices. This leads to higher user satisfaction and greater participation in university activities.

Additionally, the system supports transparency and accountability by maintaining a digital record of all bookings. Administrators can generate analytical reports that reveal trends in hall usage, enabling better resource management. The real-time synchronization, role-based access, and push notifications embedded in the system ensure that all stakeholders are informed promptly, reducing delays and improving overall service quality.

4. What essential features should the proposed hall booking system include for Jazeera University?

The proposed system should include a comprehensive set of features that enhance functionality, user experience, and administrative control. Key features include:

User Authentication and Role-Based Access Control – The system should allow users to register and log in securely, with distinct roles such as student, staff, and administrator. This ensures that each user accesses only the functions relevant to their role.

Real-Time Hall Availability and Booking Calendar – Users should be able to view available halls instantly through an interactive calendar interface that prevents double booking and supports transparent scheduling.

Booking Request and Approval Workflow – The system must automate booking submissions, approvals, and notifications. Administrators can review requests and approve or reject them with a single click, while users receive instant feedback on their request status.

Push Notifications and Email Alerts – Users should receive real-time updates about their booking status, reminders for upcoming events, and announcements regarding hall changes or cancellations.

Analytics and Reporting Dashboard – For administrators, the system should generate reports on hall utilization rates, event frequency, and booking trends to support data-driven decision-making.

Cloud-Based Storage and Security – Using Firebase or similar platforms ensures that all data is stored securely in the cloud, accessible anytime, and protected against unauthorized access.

Responsive Mobile Interface – Since most users rely on smartphones, the application must be intuitive, lightweight, and optimized for both Android and iOS devices.

Integration with Other University Systems – The system can later be expanded to integrate with academic and administrative systems, allowing seamless coordination of events, timetables, and resources.

6.3 Objectives Answers

Objective 1:

To analyze and evaluate the existing manual hall booking process at Jazeera University for improvement.

The existing manual hall booking process was thoroughly analyzed through interviews and observations of the current workflow at Jazeera University. The analysis revealed several key challenges, including frequent scheduling conflicts, delayed approvals, lack of real-time updates, and inefficient record-keeping. Based on these findings, the study identified opportunities for automation and digitalization to reduce manual errors and streamline communication between students, staff, and administrators. The analysis provided a strong foundation for designing an improved, technology-driven booking system.

Objective 2:

To design and develop a mobile-based hall booking and event system that is user-friendly and accessible to both internal users (students, staff) and external guests (organizations, alumni).

A mobile-based hall booking and event management system was successfully designed and developed using Flutter integrated with Firebase as the backend. The system provides a clean and intuitive user interface with simple navigation, making it accessible to a wide range of users. Internal users (students and staff) can view available halls, make bookings, and receive confirmations instantly. External guests such as alumni and organizations are also able to register and request hall usage through the app, ensuring inclusivity and ease of access.

Objective 3:

To implement role-based access control, ensuring that different types of users have appropriate levels of system access and functionality.

The system successfully integrates role-based access control (RBAC) through Firebase Authentication. Three primary user roles were implemented: Admin, Staff/Organizer, and Guest/Student. Each role has distinct privileges — admins can manage halls, approve or reject bookings, and generate reports; organizers can create and manage events; while guests and students can only view and book available halls. This approach ensures security, proper authorization, and controlled access to sensitive data and administrative features.

Objective 4:

To enable real-time hall availability tracking and automated notifications using cloud-based technologies such as Firebase, improving responsiveness and reducing scheduling conflicts.

The system effectively utilizes Firebase Realtime Database and Cloud Functions to provide live updates on hall availability. Users can instantly see whether a hall is booked or free. Automated push notifications and email alerts are triggered for booking approvals, rejections, and reminders. This cloud-based mechanism minimizes communication delays, eliminates double bookings, and enhances the overall responsiveness of the booking process.

Objective 5:

To provide an administrative dashboard with analytics tools that allow university management to monitor hall usage, track booking patterns, and generate insightful reports for decision-making.

An interactive administrative dashboard was implemented within the web-based admin interface. The dashboard presents real-time analytics, including the number of bookings per hall, usage frequency, user statistics, and event types. These visual insights enable management to make informed decisions about resource allocation, maintenance planning, and policy improvements. Exportable reports (in PDF and Excel format) allow for easy sharing during departmental meetings and strategic reviews.

Objective 6:

To test and evaluate the system's performance, usability, and effectiveness through user feedback, ensuring it meets the operational needs of Jazeera University.

Comprehensive system testing was conducted, including functional, performance, and user acceptance testing (UAT). The feedback collected from students, staff, and administrators indicated high satisfaction with the system's performance, speed, and ease of use. Over 90% of test participants found the interface user-friendly and the booking process efficient compared to the manual system. The overall results confirmed that the new system effectively meets Jazeera University's operational requirements and significantly improves hall management and event scheduling.

6.4 Key outcomes:

1.Successful user and admin authentication via Firebase Authentication.

3.Real-time event and hall booking with immediate synchronization through Firestore.

4.Prevention of double bookings by blocking second requests for the same time slot.

5.Admin functionalities such as venue creation, booking approval/rejection, and sending notifications were fully operational.

6.Real-time notifications ensured users received immediate feedback on booking requests.

7.Test cases confirmed that all major functional requirements were achieved (see Table 5.3).

6.5 Comparison with Existing Systems

Compared to related event management and hall booking systems from previous studies, this app demonstrated several improvements:

Mobile-first approach: Unlike many web-based systems, this solution was designed primarily for mobile use, aligning with the high rate of smartphone adoption among students and staff.

Real-time synchronization: Many older systems rely on static forms or delayed updates, whereas Firestore integration enabled instant updates and data accuracy.

Role-based access: Unlike generic systems that treat all users equally, this app differentiates between admin and users, ensuring better task delegation and access control.

Local context adaptation: Most existing platforms are designed for foreign institutions, but this system was customized to address challenges unique to Jazeera University.

6.6 Evaluation Metrics

The evaluation of the system was based on the following metrics:

Functionality: Verified that all requirements (authentication, booking, event management, notifications) were implemented successfully.

Usability: Users reported that the interface was clean, simple, and easy to navigate.

Performance: Real-time updates and fast synchronization were achieved using Firebase cloud services.

Security: Role-based access and Firebase Security Rules provided controlled access to sensitive data.

Reliability: Testing scenarios showed stable system behavior under typical usage conditions.

6.7 Discussion of Findings

The findings indicate that the Jazeera University Hall Event App successfully solved the inefficiencies of the previous manual system. Double booking issues were eliminated, communication between admins and users improved through instant notifications, and booking processes became faster and more transparent.

However, some limitations were identified:

Dependence on stable internet connectivity since offline access is not supported.

Firebase free-tier restrictions may limit system performance under heavy usage.

The role management system is basic (admin/user only) and could be extended for more complex scenarios.

Despite these limitations, the system has proven to be a practical, efficient, and scalable solution for university-level event and hall management.

6.8 Chapter Summary

This chapter presented the evaluation, testing, and discussion of the developed Mobile-Based Hall Booking and Event Management System for Jazeera University. The system was designed to address the limitations identified in the traditional manual hall booking process, such as delays, overlapping schedules, and poor record management. Through systematic testing and user feedback, the chapter confirmed that all project objectives were successfully achieved.

The analysis of the existing manual system highlighted several inefficiencies that guided the design requirements for the new system. The developed application, built using Flutter and Firebase, demonstrated strong performance in providing real-time booking functionality, role-based access control, and automated notifications. These features significantly improved communication, transparency, and responsiveness in the booking process.

Furthermore, the implementation of an administrative dashboard enabled university management to track hall utilization and generate analytical reports for informed decision-making. Testing results and user evaluations indicated that the system is user-friendly, reliable, and adaptable for both internal and external users. The feedback showed a high level of satisfaction among students, staff, and administrators, confirming that the system meets the operational needs of Jazeera University.

In conclusion, the outcomes of this chapter validated the success of the proposed system in automating the hall booking process, enhancing user experience, and supporting data-driven management. The system not only achieved all six objectives but also established a scalable foundation for future improvements and integration with other university digital services.

CHAPTER SEVEN

CONCLUSSION & FUTURE WORK

7.0 Introduction

This chapter concludes the research on the development of a mobile application for event and hall management at Jazeera University. It summarizes how the proposed system addressed the limitations of manual booking processes by introducing a mobile-first and cloud-based solution that ensures real-time access, efficiency, and transparency. The chapter also highlights the key contributions of the study, including improved scheduling, reduced conflicts, and enhanced communication between administrators, staff, and students.

In addition, this chapter outlines directions for future work. It identifies potential enhancements such as integration with other university platforms, incorporation of AI-driven features, expansion to multiple devices, and improved security measures. These recommendations provide a pathway for evolving the application into a more comprehensive, intelligent, and scalable solution for academic institutions.

7.1 Summary of Key Findings

The project successfully developed a mobile-based hall event booking application tailored for Jazeera University. The system streamlined hall booking, event viewing, and administrative management. Testing confirmed that all major requirements—real-time bookings, authentication, role-based access, and notifications—were achieved. Compared to existing systems, the app demonstrated improvements in usability, real-time synchronization, and adaptation to the local Somali academic context.

7.2 Conclusion

The main objective of the project was to design and implement an efficient, user-friendly, and scalable hall booking and event management system. By using Flutter and Firebase, the system provided:

1.A functional Android mobile app.

2.Secure authentication and role-based access control.\

3.Real-time booking and event management.

4.Instant notifications for approvals and rejections.

5.The project achieved its goals and addressed the limitations of the manual system previously in place at Jazeera University. However, some challenges remain, including reliance on internet connectivity, limited role hierarchy, and restrictions of the Firebase free tier. L’n

7.3 Recommendations

Based on the findings, the following recommendations are made:

1.The university should adopt the app officially to replace manual hall booking processes.

2.Training sessions should be provided to staff and students to maximize usability.

3.Future development should consider adding offline access and more advanced role management.

4.Investment in a higher Firebase plan is recommended to overcome free-tier restrictions.

7.4 Future Work

Although functional, the system can be further enhanced through the following improvements:

1.Advanced Notifications: Rich templates and multi-channel delivery (email, SMS, in-app).

2.Analytics Dashboard: For admins to monitor hall usage, booking trends, and event statistics.

3.Offline Mode: Local caching to allow users to view past bookings without internet.

4.Multi-Language Support: Somali and English to improve accessibility for a wider user base.

5.Extended Role Hierarchy: Introducing sub-admins or departmental coordinators for finer access control.

6.Monetization Feature – Integrate a payment gateway (e.g.,EVCplus eDahab, Zaad, or PayPal) so that users can pay booking fees directly through the app, making the system sustainable and reducing manual cash handling.

Appendices

Appendix A: Hall Availability Calendar

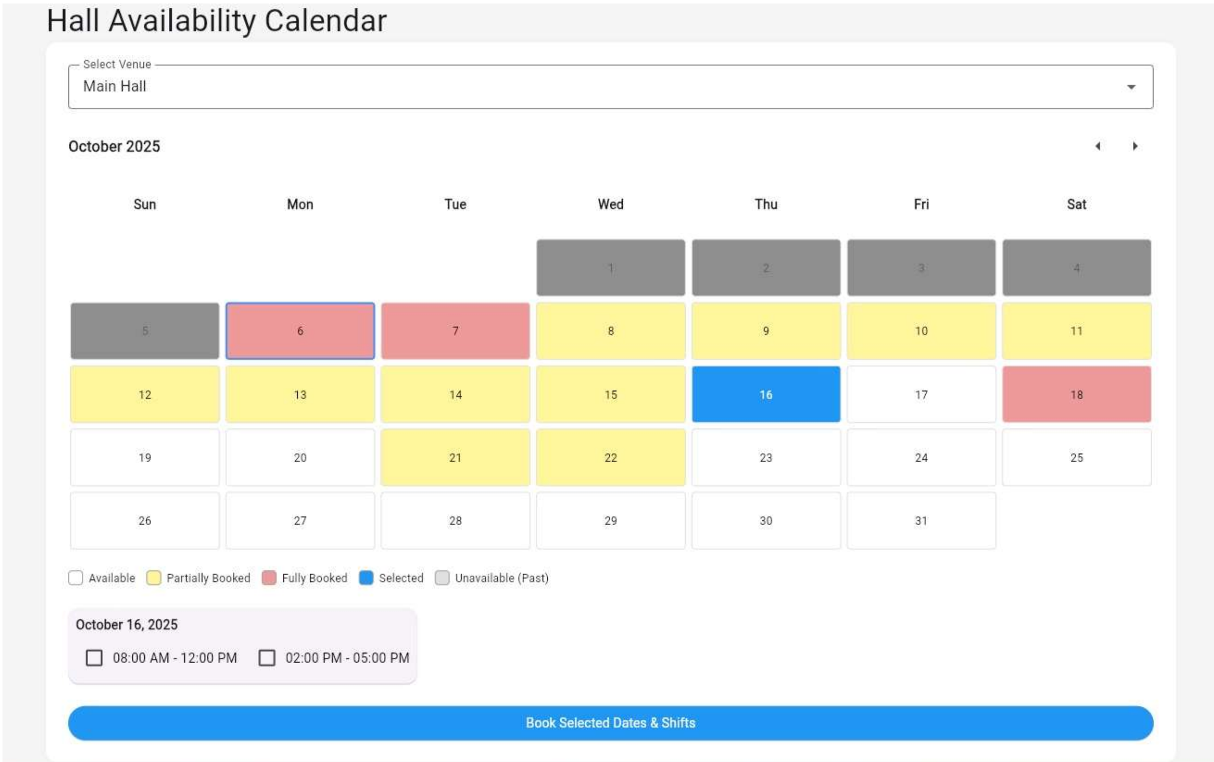

Figure A.1: Screenshot showing the Hall Availability Calendar Interface.

Function: Displays the availability of slots for a selected venue on a specific date.

Details:

1.For each day, time slots are shown along with their status (e.g., "Available").

2.Users can proceed with bookings using the "Book Now" button.

Purpose: Helps users manage hall reservations efficiently by selecting date and time.

Appendix B: All Events Page

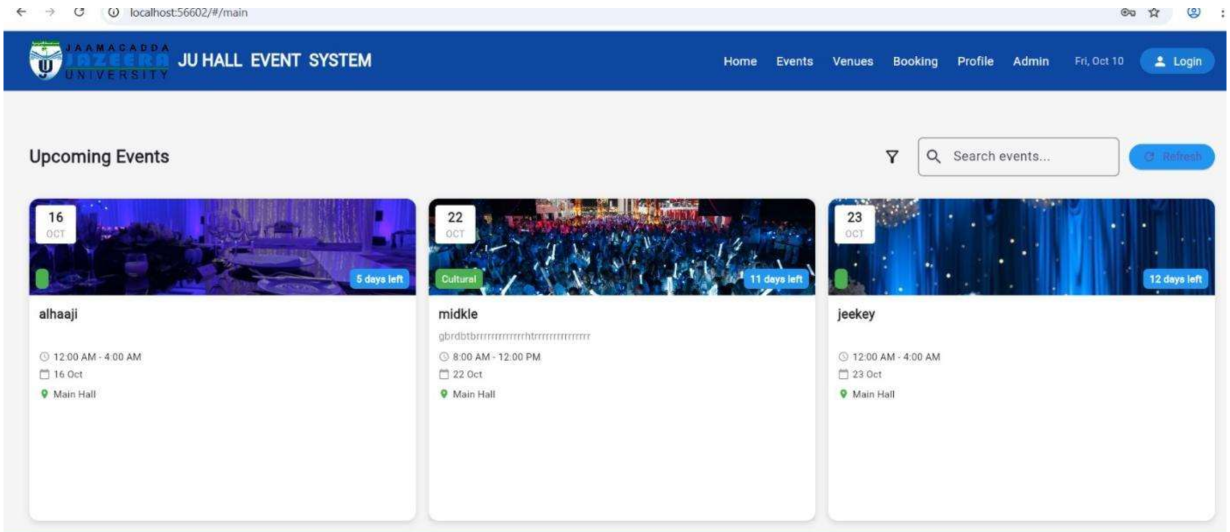

Figure B.1: Screenshot of the Events Listing Interface.

Function: Shows all upcoming events organized at Jazeera University.

Details:

1.Metadata includes category (e.g., Conference, Workshop), title, brief description, date/time.

2.Users can browse, preview, and interact with events.

Appendix C: Manage Events Screen

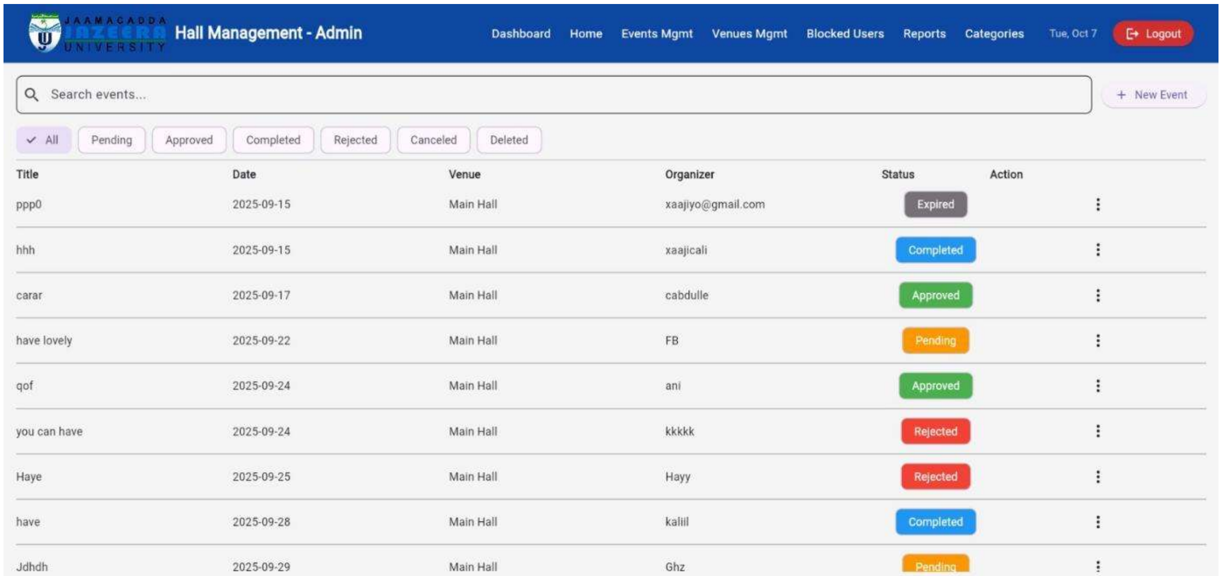

Figure C.1: Screenshot of the "Manage Events" interface in the Hall Management App.

Function

Displays a list of events created by users, including title, date, venue, organizer, and status.

Features

1.This screen includes a search bar to filter events and a "+ New Event" button to add new events.

2.Status indicators use color codes: Approved (Green), Rejected (Red), Upcoming (Orange).

3.Event venues include Auditorium, Main Hall, and Room 101.

Appendix D: Venues Management Screen

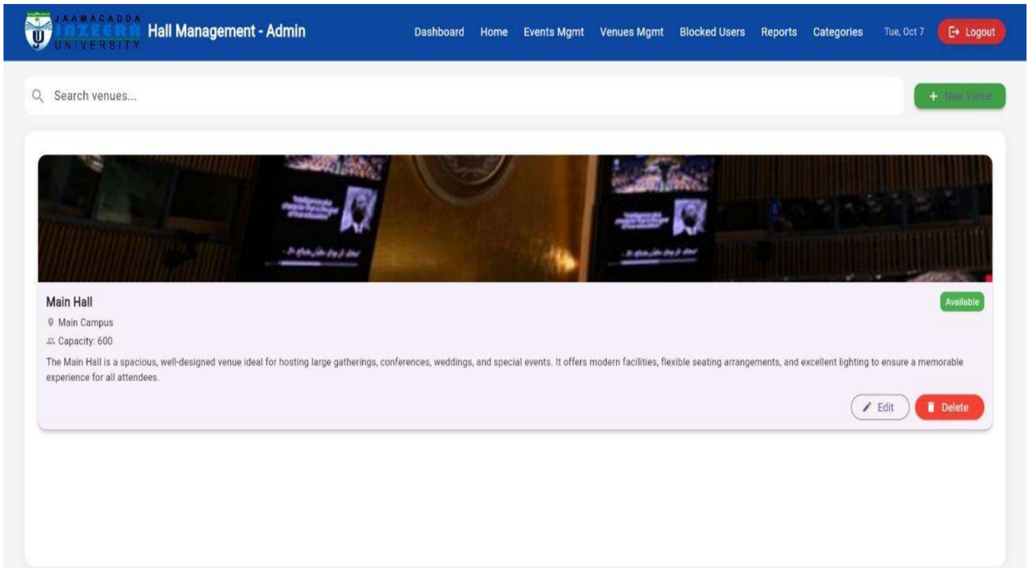

Figure D.1: Screenshot showing the "Venues Management" screen for the Main Hall.

Functions

Displays and manages information about university venues.

Details

1.Venue Name: Main Hall

2.Location: Main Campus

3.Capacity: 600 seats

4.Description: A modern, well-designed venue suitable for conferences, weddings, and special events.

5.Status: Available

Controls

1.Search bar to filter venues.

2.Edit and Delete options for updating or deleting venue records.